Geometry-Guided Identity Correction for Semi-Supervised CBCT Tooth and Root-Canal Segmentation

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Abstract. Automated tooth and root-canal delineation in cone-beam computed tomography (CBCT) requires both accurate foreground segmentation and correct anatomical identification. STSR 2026 Task 1 makes this explicit with a 65-class target and only 30 annotated volumes alongside 299 unannotated ones. We train a 3D nnU-Net on 24 annotated cases and a 40-case unannotated subset with supervised segmentation and a detached weak-to-strong consistency objective. A training-free correction then estimates a patient-specific affine-coordinate midline from predicted teeth and relabels only tooth components whose FDI laterality conflicts with their physical side, leaving root canals untouched. On the 20-case official validation set the correction raised IA from 0.2518 to 0.5493 and Challenge Score from 0.5158 to 0.7207, while image-level Dice, mIoU and normalized surface Dice stayed at 0.9563, 0.9166 and 0.9934. Memory-aware inference stayed within 8 GB and 60 s per case on an NVIDIA A10. Team `djw` placed sixth in the final Task 1 evaluation.

Keywords: Cone-beam computed tomography · Tooth segmentation · Root canal · Semi-supervised learning · FDI identification

1 Introduction

Background and Challenge Setting. Three-dimensional tooth and root-canal segmentation from cone-beam computed tomography (CBCT) supports dental planning, but must solve both low-contrast foreground segmentation and FDI identity assignment. The MICCAI STS series [10,11,12] has moved from foreground extraction to instance-level identification and joint tooth–canal analysis. STSR 2026 Task 1 provides 30 annotated and 299 unannotated CBCT volumes, requires 32 tooth and 32 canal classes, and scores overlap, identity, runtime, and memory together.

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Related Work. nnU-Net [3] self-configures preprocessing, architecture, and optimization. With sparse labels, consistency learning penalizes prediction changes across perturbed views [5], and recent STS systems add reconstruction pretraining or pseudo-label refinement [10]. Dental identity benefits from global arch and ordering constraints [6], while canal-focused and ToothFairy studies emphasize thin-structure representations [2,13].

Motivation and Contributions. Our semi-supervised model reached 0.9563 image-level Dice but only 0.2518 on the organizers' IA metric, and internal error analysis showed that many high-overlap errors were contralateral FDI assignments, making identity rather than foreground extraction the main bottleneck. We contribute (i) a 3D nnU-Net configuration combining a detached weak-to-strong consistency objective with identity-sensitive checkpoint selection; (ii) a tooth-specific correction, applied to predictions alone, that estimates an affine-coordinate left–right midline and relabels laterality-inconsistent components within the same jaw, raising official IA from 0.2518 to 0.5493; and (iii) a memory-aware inference path using slabwise resampling and blockwise argmax that keeps peak host memory under 8 GB and every validation case under 60 s on an NVIDIA A10.

2 Method

Labelled and unlabelled CBCT volumes train one 3D nnU-Net (Fig. 1); memory-aware inference produces a 65-class label map, after which prediction-only correction and cleanup revise identities without ground truth, a second model or an atlas.

Semi-Supervised Segmentation. NIfTI affines reorient every volume to LPS before nnU-Net v2 `3d_fullres` preprocessing, which handles the 17 unannotated scans stored as LAS; nnU-Net then performs non-zero cropping, isotropic resampling, intensity normalization, and patch extraction. We keep every nnU-Net default that the caution band of Sect. 3 could not separate, including cubic image interpolation, where trilinear moved canal NSD by only 0.0070. Labelled patches use deeply supervised Dice plus cross-entropy over 65 classes.

For an unannotated image x_u the network predicts a detached target from a weak view and an optimized prediction from a strongly perturbed view:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + w(t)\ \text{MSE}[\text{softmax}(p_\theta(a_s(x_u)))\,, \text{softmax}(\text{sg}(p_\theta(a_w(x_u))))]\,. \quad (1)$$

The target runs under `no_grad`, not as a moving-average teacher.[3]

[3] Blur probability 0.5, dropout $p = 0.5$, mask threshold $\tau \sim U(0.6, 0.9)$, noise from $U(-0.3, 0.3)$; $w(t) = 50\exp[-5(1-s)^2]$ over the first 20% of iterations for normalized ramp progress s and 50 afterward; the unannotated branch is included with probability rising from 0.1 to 0.5.

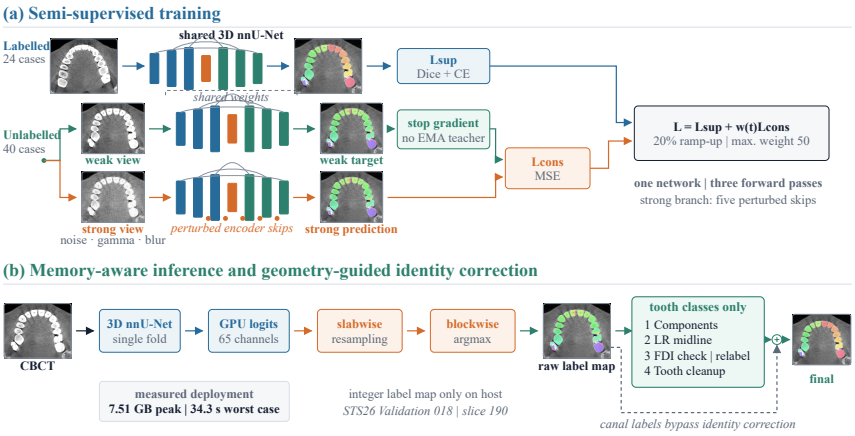


Fig. 1. Submitted pipeline. Supervised Dice–cross-entropy is combined with weak-to-strong consistency; slabwise export produces the raw label map, and affine-coordinate laterality correction and cleanup operate on teeth while canals bypass correction. The Validation 018 slice shown has 6,120 relabelled voxels and unchanged foreground.

Left–Right Flips in the Submitted Training. Because FDI classes encode laterality in the label itself, we report both places where the submitted training mirrors on the left–right axis. The supervised branch keeps nnU-Net's default augmentation: `mirror_axes` is never overridden, so all three axes stay enabled and each is flipped independently with probability 0.5, without remapping tooth class ids, which supplies contralateral supervision directly. On the unannotated branch, the weak and strong views are cropped from the same patch but their per-axis flips are drawn independently from the shared random stream, so the two views coincide on only one step in eight; the consistency term of Eq. 1 is a voxel-wise MSE with no inverse transform on the target, so on the remaining steps it matches a prediction against a mirrored copy of the same anatomy. Both paths push the network toward mirror-invariant class scores, the opposite of what FDI identification needs, and we regard them as plausible aggravating factors for the confusion corrected below. This is consistent with the ToothFairy2 CBCT challenge [1], whose winning entry reports that in jaw CBCT the cues that would let a network recover laterality under mirroring "seem to be scarce", disables mirroring on that axis alone, and finds that this "drastically improves results" [4]. We did not run a no-flip ablation—the finding post-dates the challenge deadline and the training hardware was released—so this is an uncontrolled confound, not a measured cause.

We use 40 of the 299 unannotated volumes, screened case by case for acquisition orientation and metal-artifact status rather than by a stated rule; the screening is not specified precisely enough to be reproduced, and the case list was held on the training host rather than released with the code. The selected model uses `nnUNetPlansBigMem`, generated by nnU-Net with a 12 GB planning target, giving a $160 \times 112 \times 160$ patch and batch size 2. Training runs for 1000 epochs;

the submitted checkpoint is the trainer-best model at epoch 496, confirmed with challenge metrics on the fixed internal split.

Geometry-Guided Tooth Identity Correction. Let $\hat{Y}$ be the predicted label map and $T = \{v : 1 \le \hat{Y}(v) \le 32\}$ the tooth foreground. After mapping tooth voxels to world coordinates with the input affine, the case-specific midline is $x_{\text{mid}} = \text{median}\{x(v) : v \in T\}$. Under the FDI two-digit system, quadrants 1–4 map to indices 1–8, 9–16, 17–24, and 25–32 (upper-right, upper-left, lower-left, lower-right), so classes 1–8 and 25–32 are right-sided and 9–16 and 17–24 left-sided, and contralateral pairs differ by exactly 8 within a jaw. After reorientation to LPS, $x < x_{\text{mid}}$ is treated as right. The contralateral mapping within the same jaw is $g(k) = k+8$ for $k \in \{1, \ldots, 8, 17, \ldots, 24\}$ and $g(k) = k-8$ for $k \in \{9, \ldots, 16, 25, \ldots, 32\}$. We extract each tooth class with 6-connectivity and compare every component's mean world-x with its encoded side; a conflict applies g, while exact ties and components below five voxels are left unchanged. Correction changes labels but not component masks.

Canals bypass correction because their centroids provide weak laterality cues; applying the rule to canals reduced internal canal instance Dice from 0.4512 to 0.3240. Tooth cleanup then flags duplicate classes, redundant within-quadrant coverage, and components isolated from the arch. Flagged components below 5000 voxels are removed; larger candidates are relabelled to the majority tooth class within a two-voxel dilated neighbourhood, or removed if none exists.

Memory-Aware Inference. The challenge permits 8 GB host memory and 60 s per case. We therefore retain single-fold logits on the GPU, resample in spatial slabs, apply blockwise argmax, and copy only the integer label map to the host, which kept peak host memory at 7.51 GB. The final configuration uses `tile_step_size=1.0` without test-time mirroring and was selected from runtime on the full 20-case validation sequence.

3 Experiments

Dataset and Evaluation Metrics. Task 1 contains 30 annotated, 299 unannotated, and 20 hidden-label official validation CBCT volumes. References contain background, 32 FDI tooth classes, and 32 canal classes at 0.25 or 0.30 mm isotropic spacing. With seed 42, we partition annotated data once into 24 training and six validation cases, stratified by artifact status and dentition completeness; the six held-out cases are two `_with-artifacts` volumes and four artifact-free ones, and all local selection and ablations use this split. With six cases, aggregate metric differences below $\sim$0.01 are not resolvable, so we treat 0.01 as a caution band and let no design decision rest on a difference inside it: such comparisons are settled by a second, much larger difference, by keeping the nnU-Net default, or by a non-accuracy criterion such as the hard runtime limit.

Evaluation follows the organizer implementation. Dice, mean intersection over union (mIoU), and normalized surface Dice (NSD) are reported at image

and class levels; NSD uses the challenge's 2-mm tolerance [9]. Image-level metrics binarize foreground, whereas instance metrics average class values. The identity metric is the server's `IA` field, which we refer to by that name throughout; following the series definition [10,11] it is $|\{\ell \in L_{\mathrm{gt}} : \mathrm{IoU}(\hat{Y}_\ell, Y_\ell) \geq 0.5\}|/|L_{\mathrm{gt}} \cup L_{\mathrm{pred}}|$ for the label sets $L_{\mathrm{gt}}, L_{\mathrm{pred}}$ present in reference and prediction, and internal results separate teeth and canals. The server also returns `All_Average`, the mean of these seven metrics, and a `Challenge_Score`. Its output specifies case weights of 2 for artifact cases and 1 otherwise and reports 5 of the 20 validation cases as artifact cases, but the aggregation formula is not public, so we report the returned Challenge Score rather than recomputing it.

Implementation Details. The submission uses a single fold of nnU-Net v2.8.1 `3d_fullres`, trained from random initialization on one RTX 5090 (32 GB) under Ubuntu 22.04.5 LTS, Python 3.11, PyTorch 2.4.1 and CUDA 12.1, with SGD and Nesterov momentum, initial learning rate 0.01 and polynomial decay. Inference is measured on an A10 and targets an RTX 3090, both 24 GB, alongside 8 vCPU and 29 GB of host RAM. The checkpoint has 30,836,709 parameters and 1321 GFLOPs per forward pass, and reaching epoch 496 took 12.9 h (93.7 s per epoch); these three cost figures are measured on the submitted checkpoint rather than estimated.[4] Per-case metrics were archived only for the submitted configuration, so the other rows of Table 1 are aggregates without dispersion.

4 Results and Discussion

Internal Ablations. The selected epoch-496 consistency model improves seven of the eight internal metrics over the supervised candidate (Table 1, top). The instance-level, IA, and mIoU gains span 0.0234 to 0.1039 and clear the $\sim$0.01 caution band of Sect. 3, while the two image-level changes stay inside it. These gains reflect the combined effect of the consistency objective and the larger `nnUNetPlansBigMem` patch ($160 \times 112 \times 160$ vs $128 \times 96 \times 160$). Checkpoint and objective choice matter more for canals than for tooth foreground: the epoch-1000 checkpoint loses substantial identity accuracy, and inline and offline pseudo-labeling drop canal image Dice to 0.6056 and 0.4092.

Laterality correction improves identity without altering foreground (Table 1, bottom). Correcting both teeth and canals gives the highest correction-only IA, 0.5567, but damages the canals and yields mIoU 0.5944; tooth-only correction reaches IA 0.5472 and preserves the canals while raising mIoU to 0.6262 and tooth instance Dice to 0.9077. We keep the tooth-only setting: the IA gap is inside

[4] Parameters are counted over the deduplicated module tree; summing state-dictionary tensors instead gives 88,261,509 (2.86×) because layers are registered both directly and inside containers. FLOPs cover convolution and transposed-convolution layers for one forward pass at the training patch size, batch size 1, two FLOPs per multiply–accumulate. Training time sums the checkpoint's per-epoch timestamps (wall clock 13.6 h, one interruption); they stop at epoch 496, so the full 1000-epoch cost is not logged—about 26 h at the same rate.

the caution band, the canal regression is not. The last two rows add the two changes that separate this configuration from the submitted container—tooth cleanup, which may remove or relabel tooth components, and the runtime-driven `tile_step_size=1.0`; together they move IA by −0.0045 and the eight-metric mean by −0.0029, both inside the caution band.

Table 1. Internal results on the fixed six-case split. Top: configuration comparison (supervised row on the standard plan, the rest on BigMem). Bottom: post-processing applied to the epoch-496 model of the top block's second row; the first two rows are alternative correction scopes, and the last two stack cumulatively on tooth-only correction, so the submitted row includes cleanup. Bold marks the two configurations carried forward.

	Dice_{img}		$\text{Dice}_{\text{inst}}$		NSD_{inst}			
Configuration	tooth	canal	tooth	canal	tooth	canal	IA	mIoU
Supervised, standard, ep. 1000	0.9571	0.8103	0.3632	0.4141	0.4573	0.5660	0.2392	0.3034
Consistency, ep. 496	0.9533	0.8180	0.4577	0.4512	0.5612	0.6074	0.2626	0.3577
Consistency, ep. 1000	0.9590	0.7998	0.2433	0.3151	0.3310	0.4407	0.1373	0.2068
Inline pseudo-labels	0.9351	0.6056	0.4302	0.2026	0.6252	0.2774	0.1039	0.2332
Offline FDI-aware pseudo-labels	0.9388	0.4092	0.4598	0.1148	0.5887	0.1767	0.1915	0.2379
+ teeth and canal correction	0.9533	0.8180	0.9077	0.3240	0.9564	0.4247	0.5567	0.5944
+ tooth-only correction	0.9533	0.8180	0.9077	0.4512	0.9564	0.6074	0.5472	0.6262
+ tooth cleanup	0.9543	0.8180	0.9149	0.4512	0.9652	0.6074	0.5518	0.6323
+ step 1.0 (submitted)	0.9481	0.8167	0.8977	0.4512	0.9571	0.6126	0.5427	0.6178

Official Results and Efficiency. The hidden-label validation results confirm the correction's effect (Table 2): every instance-level metric improves, IA rises from 0.2518 to 0.5493 and the Challenge Score from 0.5158 to 0.7207, while all three image-level metrics are unchanged.

Two caveats bound the row. First, it is not the submitted container: both rows of Table 2 come from stored predictions with tooth-only correction, without cleanup and at `tile_step_size=0.5`, while the container applies cleanup and step 1.0. Internally those two changes cost 0.0045 IA and 0.0029 of the eight-metric mean, far too little to explain the −0.073 difference from the final test score, but they do mean the 0.7207 row reports the correction's effect rather than the container's expected test score. Second, no component or case-level final-test results were released, so the remainder cannot be attributed directly.

Test-set composition is the one candidate we can probe internally, since artifact cases carry weight 2 and 5 of the 20 validation cases are so marked. On our six-case split the artifact cases are not the weak ones: in the submitted configuration they reach IA 0.6190 and 0.5312, against 0.4194, 0.4921, 0.5085 and 0.6863 for the four artifact-free cases. Weighting the two artifact cases by 2 therefore raises that configuration's internal IA from 0.5427 to 0.5508 rather than lowering it, so a larger artifact share is an unlikely explanation, though six cases cannot settle it. The validation result supports the correction as a failure-mode fix, but not that the same component-level gain transferred to the hidden final test set.

Table 2. Official results; dashes are metrics not released by organizers. Avg. is the server's `All_Average` field, computed at full precision, so recomputing it from the printed values gives 0.7813 for the corrected row.

Submission	Dice_{img}	mIoU_{img}	NSD_{img}	$\text{Dice}_{\text{inst}}$	$\text{mIoU}_{\text{inst}}$	NSD_{inst}	IA	Avg.	Score	Rank
Validation, raw	0.9563	0.9166	0.9934	0.4372	0.3490	0.5570	0.2518	0.6373	0.5158	–
Validation, corrected	0.9563	0.9166	0.9934	**0.6774**	**0.6069**	**0.7694**	**0.5493**	**0.7814**	**0.7207**	–
Final test, `djw`	–	–	–	–	–	–	–	–	0.647416	6

Runtime was calibrated on all 20 validation cases: step sizes 0.5, 0.75 and 1.0 left 10, 1 and 0 cases above 60 s, with worst times 150.9, 68.5 and 34.3 s, and step 1.0 cut total time from 1891 to 666 s for a 0.0064 decrease in the eight-metric mean of Table 1 (last two rows). No matched cross-GPU benchmark was run, so no RTX 3090 runtime is inferred from the A10 measurements.

Figure 2 illustrates the remaining canal weakness: tooth foreground survives in both, while thin canal branches are missed and canal classes misassigned in the failure case.

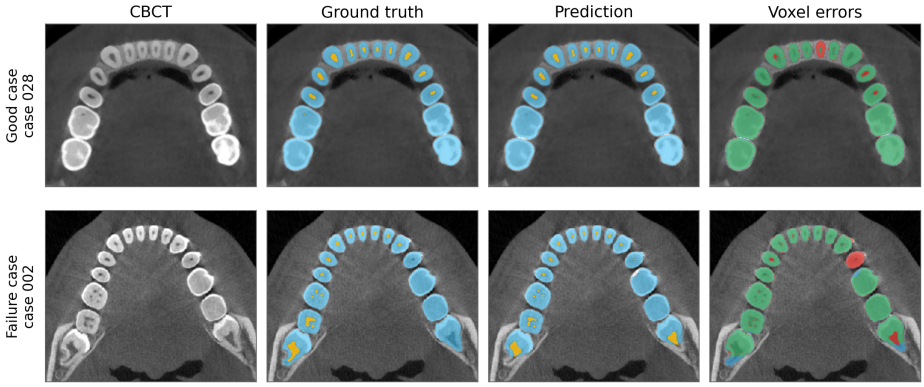


Fig. 2. Internal examples in the submitted configuration. Columns: CBCT, ground truth, prediction, voxel errors. Rows: a good case (028, slice 47; tooth/canal image Dice 0.9666/0.9162) and a canal failure case (002, slice 155; 0.9320/0.4649). Tooth/canal masks are blue/yellow; green is a correct class, cyan a missed reference, red an incorrect or additional class.

Failure Modes and Limitations. Residual errors concentrate in canal identity and boundaries: missed canal voxels go to a contralateral canal (63.4%), a tooth class (29.1%), or another canal class (6.3%) rather than to background (1.2%). The reported IA gain has a correspondingly narrow scope—it corrects tooth laterality, and the corrected validation IA of 0.5493 still leaves canals below the identity threshold—and a label audit of all 30 annotated volumes explains why inheriting the parent tooth's identity cannot repair it: of 603 reference canal instances only 53.9% lie inside the tooth named by their class and 43.8% lie inside the contralateral same-slot tooth, so no fixed offset rule does better.

The correction also inherits an untested anatomical assumption: the midline is the median world-x of predicted tooth voxels, so a markedly asymmetric dentition, or one missing many teeth on one side, can shift it far enough to relabel correct components. Our six held-out cases are stratified by dentition completeness but contain no such extreme, and sensitivity to the midline and to the remaining hand-set values (5-voxel and 5000-voxel thresholds, two-voxel dilation) is unquantified. Further limitations are the single six-case split with no cross-validation or confidence intervals, only 40/299 unannotated scans under an unreproducible screening rule, the uncontrolled comparison in which the consistency objective and the larger patch change together, absent final-test component metrics, and the left–right flip behaviour of the training setup (Sect. 2). Future work should measure a laterality-safe training run directly—disabling mirroring or remapping FDI ids under it, and aligning the weak and strong views—and learn canal-specific identity cues rather than relying on tooth-side geometry.

5 Conclusion

Foreground overlap and FDI identification must be scored separately. Correcting tooth laterality from predictions alone more than doubled official validation IA while leaving image-level metrics untouched, every case stayed inside the 60 s and 8 GB limits, and team `djw` placed sixth in Task 1. Root-canal boundaries and identity remain the main unresolved target for canal-specific modelling.

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Author Contributions Yukun Du: conceptualization, methodology, software, validation, analysis, original draft. Yude Ji: methodology, software, experiments, deployment, visualization, editing. Wei Xu: supervision, resources, project administration. Rui Zhang: supervision, resources. All authors approved the paper.

Data and Code Availability Challenge data are available from the STSR organizers under their data-use terms; the data subset and checkpoint are not redistributed. The full source implementation is available at `https://github.com/EdmundJi/stsr2026-djw.git`.

Use of AI-Assisted Technologies AI-assisted coding tools were used for the implementation and experiment scripts, and AI-assisted language tools for the English. Study design, method choice, execution, interpretation and all numbers are the authors' own and were verified by them; no AI system is an author.

Disclosure of Interests The authors have no competing interests to declare.

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